

In inventory management, these abbreviations represent critical variables used to balance the trade-off between keeping enough stock to satisfy customers and minimizing the costs of holding that stock.

The relationship between these terms is most commonly expressed in the Safety Stock Formula:

$$SS = Z * \sqrt{LT_avg \text{ times } \sigma_D^2} + (D_avg^2 \text{ times } \sigma_{LT}^2)$$

1. Service_Level_%

- **Definition:** The probability of not having a stockout during a replenishment cycle. A 95% service level means you expect to fulfill 95 out of 100 customer orders immediately from stock.
- **Usage:** Used to determine the **Z-score** (Service Factor). Higher service levels require exponentially more safety stock.
- **Function:** Acts as a strategic target to balance customer satisfaction against inventory carrying costs.

2. Avg_Daily_Demand (D_avg)

- **Definition:** The average number of units sold or used per day over a specific period.
- **Usage:** Forms the "baseline" for replenishment and is a primary multiplier for lead time risk.
- **Function:** Tells you how much stock you *expect* to lose daily under normal conditions.

3. Demand_Std_Dev (sigma_D)

- **Definition:** A measure of the "swing" or volatility in daily demand. A high sigma_D means sales are unpredictable (e.g., some days you sell 5 units, others you sell 500).
- **Usage:** Used to calculate the "Demand Uncertainty" portion of safety stock.
- **Function:** Triggers the need for extra buffer stock to protect against unexpected sales spikes.

4. Avg_Lead_Time (LT_avg_days)

- **Definition:** The average number of days between placing an order with a supplier and receiving the goods in your warehouse.
- **Usage:** Determines the "Reorder Point" (ROP).
- **Function:** Defines the "window of risk"—the period during which you are vulnerable to stockouts because you cannot get more inventory quickly.

5. LT_Std_Dev (sigma_LT_days)

- **Definition:** The variability in supplier delivery times. A high sigma_LT means a supplier is inconsistent (e.g., they usually take 10 days but sometimes take 20).

- **Usage:** Multiplied by D_{avg}^2 in the safety stock formula.
- **Function:** Protects against "Supply Uncertainty." **Note:** This often has a much larger impact on safety stock levels than demand variability.

6. Safety_Stock_Units_AI

- **Definition:** The "extra" units held in inventory to protect against fluctuations in demand or supply. The "AI" suffix often refers to **Artificial Intelligence-optimized** levels, where machine learning predicts these buffers more accurately than static formulas.
- **Usage:** Added to the cycle stock to determine the total inventory level.
- **Function:** Acts as a "shock absorber" for the supply chain.

7. Inventory_Value_

- **Definition:** The total monetary worth of the stock held, usually calculated as: Units times Unit Cost.
- **Usage:** Reported on balance sheets as a current asset.
- **Function:** Measures the "Capital Tied Up." It helps managers understand the financial risk and the "holding cost" (insurance, warehouse space, and opportunity cost) of the inventory.